

ActInf GuestStream 054.1 ~ A Gómez-Emilsson & C Percy ~ Electromagnetic

GuestStream | A Gómez-Emilsson & C Percy ~ Electromagnetic Field Topology Consciousness | 1:19:40

Hello and welcome everyone. This is Active Inference live stream, guest stream number 54.1. It's August 25th, 2023. Today we're going to be hearing about electromagnetic field topology as a solution to the boundary problem of consciousness. This is going to be a co-presentation followed by a discussion with our guests here. So I will leave it to everyone to introduce themselves and present and discuss however they prefer. And thank you again for joining, please.

Thank you so much for having us, Daniel. Yeah, I'm Andres, co-founder of Qualia Research Institute. And yeah, I'm here with Chris and Libor. Chris, co-author of this paper, and Libor, a volunteer here and Wiki Extraordinaire for QRI. But yeah, go ahead, Chris and Libor. Yeah, no, no, no more to say that. That's plenty. Let's get into the detail.

Yeah, I'm co-working with QRI and a lot into Active Inference. So I'm trying to unify QRI with Active Inference a lot. And yeah.

Beautiful. Yeah, I mean, essentially, yeah, next, next slide, essentially, our outline of this presentation is going to essentially walk you through some of the key ideas of the paper, and kind of like trying to illustrate them or making them a little bit more intuitive, fairly high level, although we will actually present kind of the core argument. The main thing to point out here, next slide is that the theories of consciousness, you know, I guess, like, 20 years ago, they sounded like a total, total crackpot type of of the fact that the theory of consciousness is going to be a little bit more.

Yeah, I've still got a good connection. So we can give them a moment or two. Yeah, if not, I'll pick up from what was the crackpot?

If he reconnects.

Oh, my gosh.

Sorry, Andres, please repeat from crackpot.

Oh, yeah. Oh, my God. Yeah. Over the last 20 years, however, there's been an explosion of interest in electromagnetic theories of consciousness and serious peer reviewed publications about them, especially the last 10 and then five years, kind of like a variable exponential explosion in terms of a number of like serious articles published. And one of the very, very kind of like key amazing kind of like contributions that something like an electromagnetic theory of consciousness can offer to the overall ecosystem of philosophy of mind is that it provides a way to solve question for how information can be combined. I mean, essentially, when you have information distributed, how is it possible that it can be simultaneously part of some kind of like unified whole and electromagnetic theories of consciousness are just very proud to say, well, the electromagnetic field is ontologically unitary, it's already unified.

So there's not much of a problem here. However, you know, once you kind of like have this this unity, then the question becomes, well, but how do you actually draw a boundary around it? So actually, you just traded one problem for another. So yeah, dig into the details there.

Cool. Shall I take us through the literature review as context for this? And then Andreas will come back to kind of give you the solution that we think can fit with electromagnetic fields. So yeah, the boundary problem, let's say so binding problem you heard just now into a first approximation that is, whatever you think might make up consciousness, maybe it's neurons firing, maybe it's some sort of magical qualia at the bottom of the universe, whatever you think it is, these small things need to be bound together to reflect the complex phenomenology we have, we see lots of things all at once right now, I'm, I'm smelling the flowers outside, I've got the laptop screen in front of me, there's lots of stuff happening all at once in a unified way. So that's the binding problem, we won't do more on that here, we'll sort of assume some familiarity with it. But then the boundary problem is, well, once you've got a mechanism that creates the binding between these lower level entities, why would that mechanism ever stop? Because clearly our awareness, our phenomenology doesn't go on forever. We don't know what's happening in Alpha Centauri, our awareness is not unbounded. And as soon as you realise that you need boundaries around something, a number of things suddenly don't quite make sense. So, Owen Schrodinger of physics fame flagged this up as one of the things that he felt was completely unresolved. So despite his, his kind of vast intellect and his contributions elsewhere, he felt that this puzzle was one of the things that he described as when he turns his mind to it, everything just evaporates, none of it, none of it makes sense anymore, he can't get a grip on the problem. And he described the problem. As you can see on the screen, he was saying, why is it that my identity is at the intermediate level, this kind of roughly human shaped body. And you can say, okay, we mostly mean the central nervous system. But either way, this, this broad entity has lots of smaller things inside it, cells, organs, human bodies, and lots of things above it, cities, states, solar systems, universes, and so on. Why is our self consciousness always at this, always at this one level? And if you describe that as a problem to somebody who's never heard of it before, or never thought of it before, they may, this happened to me just, just last Friday, they might say, well, it's obviously just like, the body stops, like the skull stops here, or like my arm stops here, like, of course, it's not going out 12th century, but why would it? And that's, it's a very intuitive, immediate human reaction. But you probe it a little bit as Schrodinger did, and it kind of falls apart. Because if you're a neuron firing somewhere in the central nervous system, and you're being initiated by a neuron in the peripheral nervous system, how do you know to stop binding? How do you know that there's a boundary that stops at the edge of the central nervous system? Or if you're, if your fingers are holding onto a tool, and you're getting feedback from that tool, you're driving a car, it can feel like you're part of that car. How does, where does the system know where the boundary is? Because most of these things that we observe, have only observer dependent boundaries. If you've got a wheel in a car, rolling along the road, from one perspective, it looks like the wheel is separate from the road. But if you're a tiny ant or something, and the car is parked, these things all may feel like they're one object. And this was when when Schrodinger turned his mind to this, it puzzled him. So there's that that's given the benefit of the doubt, let's let's say that there's something real to be puzzled through here. This issue applies both to the example I started with around building up

in small units. But also, if you're a cosmopsychist, you believe that there is one universal consciousness, and we are like a god playing with its puppets. In that case, you still have the same question, because you're still demarcating boundaries, you just get the mirror image, we'll mostly focus on the idea of expanding up from small things, because that's closer to the orthodoxy. But there's no, you can't solve this problem magically by declaring a single consciousness. Okay, so that's, that's what that Schrodinger didn't say much more about this, he just said that this was a, this was a puzzle. The first person we found that really formalized this and gave it the name boundary problem was Rosenberg. And it was a number of pieces late 90s, early 2000s. And he uses the phrase Scylla and Charybdis. But it's exactly the same idea as Schrodinger had. One of them is, well, why aren't there lots of small things? I think that's, I think that might have been the Scylla that might have been the nine headed monster. And then on the other hand, what about the things that we are nested within, and he called that the charybdis. And he said, it's very hard to posit a genuinely robust mechanism that would stop it in both directions. You can imagine a reason why lots of small things get bound together, but why would the binding stop? And his view in framing this problem is that no one had solved it. Winding on to Andres and I chatting about this last year and building on some of QRIs foundational work in this area, we wanted to know, well, what's happened since Rosenberg framed this and Schrodinger was so puzzled by it, like, surely someone's come up with a few solutions to it. So cue standards, database driven literature review. And you've got the details there, fairly standard stuff. What we what we looked looking across the Scopus database, we only found five papers that seem to be relevant about the boundary problem, compared to 92 on the binding, and 47 on the combination problem. So some of these related problems. And in our intro, we kind of went straight from binding to boundary, they are so you know, if you're thinking about one, you really should be thinking about the other. Well, that is not reflected in the literature, we did a bit more puzzling around. People don't seem to be referring to this by other names, or when they are, they haven't solved it either. So it felt that there was something worth going at here. And this is what what initiated the paper. What we did find, though, in partly in those five papers, and partly in looking around, we do find families of possible solutions. And there's no clear winner, but maybe philosophy rarely has a clear winner for anything. So maybe that's an unreasonable expectation. But there's definitely open questions for each class of theories that have been put out there to solve this problem. So the first of these loosely termed algorithmic, so computational causal structure theories, I'm just trying to summarize them into one term, some of the examples there, IIT, of course, very well known, global workspace theory as well. They will tend to say that things bind together when they're in a particular pattern. So you might imagine a database that references itself, or the working memory of a computer, or as soon as things are somehow linked together in a causal sense, in IIT, there's a particular type of pattern. But nonetheless, it's an abstract informational relationship. Those papers don't tend to be super explicit about defining boundaries. Normally, they say IIT will normally say, here's our system. And now let's ask within that system, where is the

highest ϕ ?

But they sort of just say, take the boundaries on that. So no one has said, let's take the whole universe as a system that the math is already virtually impossible at even toy systems. But reading between the lines or reading some of the other papers around it, there's something about if you had a high ϕ area surrounded by lots of low ϕ areas, well, it's natural that it's the bit that's high. And there's this principle of exclusion that says you take the highest one in a system. Maybe you have to specify a system before it makes sense to ask that. And if you took the highest bit out, somehow you just remove some of those causal interconnections that create that that highness, whatever's left will have a slightly lower bit of ϕ . And that is now where the sort of locus of consciousness resides. This may well work, kind of the maths would need working out. I think Andres and I are initially skeptical. But because as best as we can see, no one's really created complicated maps and discussed what would happen when you've got different mixtures

of densities that are shifting over time, what stops it from being arbitrary. Certainly, there are there are questions to be had, and we can come back to this a bit if we want to talk about it further. But suffice it to say, it's not a guaranteed solution. And there's one critique that we cite there. Resonance theories.

And a link between those and electromagnetic fields now as well in that the main progenitor of these theories, Tam Hunt would say that the main resonance, which he sees binding happening together in human consciousness is EM fields. So there are links between these as well. And maybe the informational patterns are being encoded in resonance, or quantum entanglement, or EM fields. So there can be links between these as well. The resonance theory that says the binding happens when things are in the same frequency, simple enough, or when they are close enough, and that's becomes a little bit more debatable. But there is, you know, what does that really mean? The boundaries, there is a conceptual solution for this, which is the slower shared resonance in a system. But many of the same issues applies with as with the algorithmic. But again, if experimenters, it may be possible to create, create one that works there. Quantum theories, entanglement is what binds it together, at least in some of them. So what you know, what causes two experiences, a blue cup, and the shape of the cup to be bound together into one? Well, it's that the blueness and the cupness, I'm obviously talking at a very high level of abstraction, but those are somehow quantum entangled with each other. That would indeed create, to the best of our physics knowledge today, a kind of ontologically real binding. And that's that's quite an attractive part of the solution. Well, what would stop those boundaries? There is the monogamy principle, things can only be maximally entangled up to a certain point. But there's a very very open question about whether that sort of thing would would enable the kind of macro phenomenology we

have. But in the paper, we say, all of these are worth exploring, like, let's, let's keep investigating.

And then there's the one I started with an introduction, like maybe it's the boundary of the central nervous system, but hopefully even a small amount of scrutiny causes that one to collapse.

And most of these explicitly say that more work is needed, more maths is needed. And certainly that will be true of what we're presenting here as well. What we want to do is add a new one to this list.

And the inspiration for this is that, well, if the EM field accounts present a solution to the binding problem, they certainly believe they do, there's lots of literature on it, then maybe they can also provide a solution to the boundary problem, given how closely related they are. When you read those accounts, they don't quite have it nailed yet. The most common explanation they give is a kind of inverse square law rule where, well, the strength of the field just becomes so small, so quickly, you know, by the time you're a centimeter out from the thalamus, or by the time you're at the edges of the skull, these fields have just evaporated into, they decohered, they're indistinguishable from the neural noise all around us. That might be okay, if you're okay with a fuzzy boundary, and we'll come on to that in a sec. I always motivate a phase transition a bit like quantum entanglement, or a bit like the exact resonance of resonance theories. But those phase transitions are pretty speculative as well. So, the solution we will come to, and Andrew's will outline, is a different aspect of EM field. It's not fuzzy boundaries, and it's not a vague phase transition. It's an entirely different topological solution, which we argue can solve the problems. What we think is worth doing, though, is to make that boundary problem more explicit. What you've heard both from Schrodinger and from Rosenberg is this idea of, kind of like, why aren't the cells inside you conscious? Or if they are conscious, why don't we ever experience that? And what, again, going up, what about cities and families and things, or group chants, or football stadiums, that sort of stuff. And there is a scholar who argues, based on resonance theory, that actually, when we are in a group chanting or a group dancing situation, our consciousness does, in fact, move to that level as well. And that's, it's speculative, but you know, there are, there's much to be thought about. So, to make this problem more precise, we've got five. And I'll just give you the one or two sentences on each. The hard boundary problem is this sense that a fuzzy boundary probably isn't good enough. And let's kind of explain a bit more why. If you have something like a wave, or the example of the wheel I gave before touching the ground, or indeed any connection between any physical atoms, the closer you look to it, the less obvious it is where it stops. The boundaries of my skull may feel very precise to me, but if there were an alien the size of a solar system, their ability to perceive this would most likely just blurs imperceptibly into its surroundings. Like these things look, feel hard, but they're not really hard.

David Pérez- At the edges of our awareness, there's this sense that things like, might feel fuzzy, but at some point, it just sort of stops, might feel like it, it might feel infinite. There are states of mind where we can, we can feel an infinite awareness. But probing that further, it doesn't truly encompass everything and go on forever. At this edge, our assertion is that sometimes this feels like a hard boundary. And if it ever sometimes feels like a hard boundary, then you need some mechanism that is capable of a hard edge, in a way that these physical phenomena typically aren't. So that's the hard boundary, definitely happy to like drill into more of any of this or any of the other five, if they're not making enough sense in the Q&A. So lower levels, we've covered that. That's, I mean, the great example of this is, we know that when we go to sleep, some part of our consciousness, at least the bit that we remember switches off, yet the brain remains active. The lower levels boundary problem could be reinterpreted as saying, why isn't, why doesn't our awareness just shift to another part of the brain at

that point? It certainly doesn't seem to, most of the time. Higher level boundary problem. We've had that from the beginning, that's why doesn't it go up? Maybe it does sometimes, certainly doesn't normally. The private boundary problem. So this is now rather than going down into ourselves or up into the universe in which we are nested, this is more take two of those boundaries, me and you, me and one other person. Why can we

not bridge into each other's minds, at least not routinely? Those who claim to do so have struggled to replicate it and get it into the mainstream. So this is privacy. There appears to be some limitation that stops these mechanisms from connecting with each other. And yet, if it is slower shared resonance, or if it is highest informational phi in IIT, why is it not possible to construct a system in which the highest phi is now an interconnected one and our consciousness is collapsed? Maybe it is possible, and that's just a technical endeavour we haven't yet achieved. But either way, there's something that's challenging about it, either as a technical barrier or as a philosophical one. The last one is the temporal boundary problem, because everything we've said so far is in space. But the kind of entities and experiences we're talking about are not zero time space-based activities. They have a temporal persistence. So what is it that puts boundaries around a micro moment of experience? What is it that enables us to experience the end of a note and still retain that idea of the start of the notes and experience them together? And how does this extend through broader experiences of the self? So those are the

five puzzles, hopefully, that's enough of a qualitative overview, so that you know what we're trying to solve at least. And if I hand back over to Andrews now, he will give you an entirely new solution to these puzzles.

Andrew Sarkozy

Yes. Next slide, going a little bit, yeah, kind of like a bit broader, too, in that what Chris just talked about is kind of the criteria for the boundary problem, kind of like breaking down the boundary problem into subcomponents. And then there's like also like more broadly, like things that you would want out of a theory of consciousness. And so like the solution to the boundary problem ideally will also in a sense like satisfy these constraints. And these constraints are frame invariance, non-nipi-phenomenalism, and no strong emergence. Next slide. So frame invariance essentially can be recolorance invariance. The fact that like, whatever structure is giving rise to the boundary shouldn't, in a way, be frame specific, shouldn't just depend on like how you're looking at an object, it should be kind of like an intrinsic objective property of the object, that if you move it around, it shouldn't in principle, you know, be changing in that way. Next slide.

Non-nipi-phenomenalism, for those who are not familiar with the term, this is the idea that for a particular conscious physical system, the corresponds and experience, but that there's no causal influence from that conscious experience back into the system. And there's like various ways of kind of trying to ground the question of, hey, why is epiphenomenalism not the case?

There's kind of like, yeah, differences in emphasis by each of these questions, but they're roughly pointing in the same direction. Like one important question is, how is it possible that we can talk in

detail about our experience? But the way we actually framed it in the paper that we find the most compelling is, how was evolution able to recruit, you know, coherent, globally bound states, even if they're not causally significant, right? Like if they're not causally useful, it would just be an extraordinary coincidence that it just happens to happen simultaneously, along with something that seems to be computationally significant and, you know, encompass a whole lot of information at the same time. Next slide. And then we also have no strong emergence. There's this quote from David that says the Lagrangian of the standard model does not break down inside one's own skull. Meaning we want the laws of physics to apply universally to whatever explanation space we come up with, and we don't want to introduce special cases or say like, oh, there's like a special different field in the brain or anything of the sort. We want to have homogeneous set of physics that, you know, as simple as it gets, as mainstream as it gets in terms of a feeling to standard physics. Next slide. So yeah, the overarching solution and explanation space that thing is electromagnetic field topology. So first of all, I will walk you through in what way this explanation space satisfies the kind of like three constraints that I provided. And then Chris is going to explain how it addresses each of the sub problems. So next slide. So yeah, next one. Yeah, if you actually kind of look at these amazing reconstructions of the electromagnetic field of stellar bodies, you would see that's nothing trivial about it, right? Like there's like all of these incredible complexity and richness in structure. A lot of it is very counterintuitive, you know, not something that you would see with the naked eye, you see kind of this, you require superimposing a lot of pictures from the different regions of the spectrum to kind of like get a picture of what's going on in the electromagnetic field. And yeah, it has kind of this elongated weird plasma tubes and, you know, circular structures and loops and so on. So it's a, you know, a very structurally rich environment. Next slide. And, you know, the way in which, you know, something like the topology of that field will satisfy the criteria that the weight line is as follows. So if you look at the electromagnetic field of the sun, sometimes this phenomena called magnetic reconnection happens. And this is actually responsible for solar flares and kernel mass ejections. It has to do with when plasma tubes that are the electromagnetic confinement structures touch with each other. And so they can, you know, open up and spray plasma into the outer space. The key thing to point about here is that this is actually a topological change to the field is not just any kind of physical change. You know, when you have a, you know, solar flare, coronal mass ejection, that is the effect of a top, but macroscopic gigantic topological transformation. What this slide is trying to get at is that whether a system experienced magnetic reconnection or not, will not depend on your frame of reference. And the very simple reason is that Lorentz transforms simply squeeze and stretch, but they never actually, you know, reconnect or cut or, you know, add any kind of like topological change. In other words, a Lorentz transform will leave a topological event untouched. So in that sense, okay, like topology as an explanation space for why, why this is frame invariant, it definitely fits the bill. Next slide.

I importantly to you in the brain, you know, this, we also can tackle the question of like, in what way the field is not epiphenomenal. There's a lot of research on local field potentials in the which are, yeah, little microscopic electric oscillations. And they do seem to actually have

causal effects on the activity of neurons. In other words, people might believe in a sense that, hey, brain waves are kind of just a reflection of neural activity. But the thing that something like transcranial magnetic stimulation makes clear is that on top of that, they actually feed back into the probability for neurons to fire, especially the probability for neurons to fire in synchrony or in a state of coherence. In other words, the actual causal structure of the electromagnetic field and how it connects to neural activity is this dual causal arrow where neural activity influences local field potentials, neural activity influences activity via synapse.

Yeah, we've been the victim of the connection again. Let's hope it comes back as quickly as it did last time.

Hmm. Yep.

There's a lot to say, but I won't.

Hopefully you got some, some harsh questions written down as we've been, we've been taking you through it.

Yeah. Just enough people who are listening in the discussion, we'll look at some questions. So please feel free to write anything.

Personally, I'm, I'm curious whether the awareness is going to be the light of the sun, the heat of the sun, the ejection, the loop structure, the birth and death of the star.

Yeah. Yeah. I mean, if there's, if the idea that you'll, you'll see here stands, then you've got the idea that a, a closed topological pocket of electromagnetic field, that's, that's where awareness can take place. That's where a first person perspective, that kind of a locus can exist.

But what it's aware of, well, that's an entirely different question. And if there are pockets like that, I mean, I, we're, we're not asserting in this paper that the sun has these, it's such a complex entity.

It's very possible, but that's, you know, that is a different, but if to the extent they are there, what on earth would be experienced within it? That's a, that's a different thing.

The content problem. It's not just for YouTubers. It's also for conscious systems, but yeah, you've bounded all together.

You've delimited a boundary topologically and then about what? And so what?

Yeah. Yeah.

Let me, let me pick up a bit while we're, while we're waiting for him to come in. So yeah, this is, and there's really growing kind of neuroscience and biological evidence for downward causation from electromagnetic fields in the brain.

And all we really do in this paper is acknowledge that research literature exists and say, and therefore it's possible for fields to be non-epiphenomenal. We don't get into the specifics of it, but they've been shown in memory formation.

They've been shown in the olfactory bulb for your smells. They've appeared all over the place. So there's a, ah, Andreas, you're back.

It's just making the case for non-epiphenomenal fields in the brain.

Oh, fantastic. Beautiful. Yeah. Sorry about that. Hopefully it doesn't happen again. Okay. Next slide.

Yeah. And, uh, I mean, think about like the immense causal power that a topological change in the field has in cosmological terms, you know, like this billions of tons of plasma being ejected just because of a, you

know, otherwise like epiphenomenal magnetic reconnection is like, no, no, actually, you know, topological transformation.

Yeah. And so, uh, really, really significant at a massive, massive scale. Uh, next slide.

Yeah. Uh, next one. Yeah. Just showing kind of how ridiculously powerful this can be.

Yeah. And, uh, importantly too, um, the actual fact that you get an enclosure with a topological transformation in a medium, uh, also gives you this capacity to essentially have,

holistic behavior. Uh, this is one of the things that to me was kind of the most exciting about this explanation space.

Like the moment you can close the field lines or, you know, do a top formation waves will be confined.

Some kinds of waves can be confined within a topological pocket.

So, uh, when you have confinement of waves, then you have resonant modes. And when you have resonant modes, then you have this very fantastic effect where the whole shape of an object essentially is expressed in the way every part of the object is vibrating.

In, in, in, in, in which case, yeah, essentially you have this reason why evolution would recruit, you know, these topological objects is because it actually allows organisms to behave as a unit.

So I think, yeah, that might be a key piece of the puzzle for IP phenomenon.

Uh, next slide. Uh, these are just like, yeah, kind of like some of sketches of some of the kinds of boundaries that in principle you may have in the electromagnetic field and what kind of, uh, you know, binding in consciousness they might correspond to.

Um, but you know, you don't have to, you know, just speculate to the really exciting thing here is that, uh, you can actually play with simulations of the electromagnetic field to try to make sense of the topological transformations that happen in it.

Uh, so the next following slides, uh, we can just go through them fairly quickly are essentially resonant modes of an empty cavity, electromagnetic resonance modes of an empty cube cavity.

Where the left side is the electric field and the right side is the magnetic field.

Um, the website where you can check this is, uh, called filestat.com slash EM box.

And, uh, the thing that it kind of like really neatly shows is how the magnetic field sometimes get this, uh, close looping structure.

Uh, next slide. Yeah. Just go through, through them, maybe like a five seconds for each of them.

Um, and, uh, it also allows you to essentially look at what happens when you stack some of these resonant modes and some combinations, linear combinations of these resonant modes create intricate edding, edding, eddings and loops and twists that loop around and create this topological boundary.

Very fleeting. It has to do with like the, you know, the frequency of oscillation of the electric field.

You know, how many times per second you're going to get this looping structure.

But the fact that, yeah, you can show there's some topologically closed objects, uh, that appear there is quite remarkable.

And overall the idea here would be that, well, something like this is happening in the brain and the brain is actually recruiting these, uh, looping architectures.

All right. I'll hand it over to Chris.

Cool. Thank you very much.

Yes. We've seen that electromagnetic fields can form closed loops and people have used that some of the sites in the paper around people trapping knots of lights.

And it's just kind of really remarkable things happening at the edge of, um, uh, sort of room temperature state physics, really incredible stuff.

We know that in principle, these fields are capable of downward causation, um, in massive contexts like the sun, but also in very real human brain settings.

So conceptually, and we know that topologies in general are frame invariant.

So we've got, we've got those things in place.

We've got boundaries that satisfy those, those various features that we wanted.

We don't know exactly how they occur in the human brain.

All we're presenting is a conceptual solution that those things are possible.

But what we do say is, well, what would, what would such a solution look like to those five, uh, boundary, uh, problems that we just described.

So taking them one by one, um, or rather, yeah, taking them as a, as a group.

Let's first say what's in such a theory, the first person perspective, that kind of phenomenal awareness, the, the, what it's likeness that we experience in a very normal everyday sense.

What could that reflect?

What could that correspond to in such a model?

And there are others, but we'll give one example and use that to walk through the problems.

So we would say that once you've got a closed topology and it may well require the time dimension for a time.

The time dimension for it to be properly ontologically closed.

And at that point, EM energy of that specific spectrum, same spectrum that's making up the boundary can't pass through it.

And that's, that's the whole point about the closed loop.

Other things might pass through it.

Sound wave might pass through it, but nonetheless, for the entities that we're interested in, it is closed to itself.

And that's where we get that.

That's where we can get this hard boundary.

There may be lots of these being created in the brain.

The brain is absolutely chock-a-block with different bits of EM generating, um, underlying phenomena.

So it's certainly plausible that some of them are closed.

We don't know.

For this to be true, there must be at least one, but it's plausible that if there's one, there's probably several.

So you've got several of these.

By definition, they are non-overlapping.

They, the boundaries cannot overlap, but they could well be multiple at the same time.

One at the front, one at the back, could be all over the place.

It's probable though, that only a subset would be continually recreated in a stable manner, because the fact

that these brain, these fields can have downward causation, they can do useful things.

Some of the sites in the paper, um, about information processing point to some of the non-linearities you can achieve through wave and field-based resident mode based, uh,

uh, computations, which may well have benefits from, from an information processing perspective.

So in some cases, these will be continually recreated.

Only one of them, because they cannot overlap, only one of them can enclose, let's call it an immediate memory model, as a conceptual point, not, not committing to what exactly that might mean.

Uh, we suspect these wouldn't last very long and they probably don't extend over the whole brain.

But who knows?

Regardless, it would vary from period to period.

Other field theories might have other ways of recruiting this concept for solving the boundary problem.

But let's walk this one through and see what it looks like as a response to those five things.

So what we're saying is this closed topological pocket that is kind of axiomatically the same as a first person awareness.

When you've got that, you've got something that is a, is capable of both binding together those micro bits of experience and presenting them within a boundary.

So they have some unified ontology.

So one by one, and I think this is our last slide.

The hard boundary, well, that comes from what we've been talking about.

Energy of the same spectra can't pass through that, the closed loop.

So that, that happens quite naturally, a bit like with the field solution to binding.

This is something you almost get for free as a result of using that.

Lower levels and higher levels.

Well, because only one of them, because of the non overlapping point, because only one of them can include the immediate memory module, which is crucial for the next bits.

Then you can't have lower levels and higher levels.

There may well be mind dust at those levels, which is this kind of lovely evocative concept.

There may well be flashes of tiny bits of awareness that's kind of, they never repeat.

The pattern's just gone.

There's nothing that links it into future versions of itself.

Just gone.

The universe may be replete with these things.

It's certainly got plenty of electrons in it.

Why not plenty of these things?

Who knows?

But there's only one closed pocket that, maybe I'll come back to four, that can enclose this immediate memory module.

And that's crucial because it's what creates a temporal boundary that in any possible sense could reflect what we experience day to day.

Because these tiny, if you think of the, these 4D pockets, if one pocket were to last long enough to encompass, say, a whole day's worth of your experience, it's almost impossible for it to vary enough without breaking to actually contain all the types of complex phenomena we experience during the day.

Which is why we think it's much more likely they persist for a short period of time.

But nonetheless, our experiences of things persisting for longer.

Now the micro level persistence, why is it non-zero?

Because if something is infinitesimally small, it probably doesn't help us.

If something is the equivalent of a geometric point, no matter how many of them you put together in a physical setting, you're not going to create a line.

We want some non-zero duration.

Well, we've got that for free because the closed pocket has this 4D ontological presence.

That's fine.

How would you get, but it's, but it may be really, really short, far shorter than we experience even this sentence.

So what ties that together?

Well, because it encloses each millisecond to millisecond, it recreates.

Each one is a different piece of awareness.

They are not in any way ontologically linked to each other.

They are each separate.

But because each one encloses the memory module, which is just an information processing thing, you can encode memory in pen and paper.

There's nothing magical about this.

By referencing it, it knows what was just there before.

And that can create, you can, it's maybe, we could call it an illusion if you want, whether that matters is a, is a much more philosophical point.

But you create this pseudo time arrow.

You have unified insights because that's what fields do.

They unify the insights in all the things that make it up.

No information is lost.

It is all bound together.

So now you've got the recent pasts and then you've got the bit before that from the previous pocket and a bit before that and a bit before that.

So it's all bound into each, each individual presence.

As short as it may be, it has the sense that it's something bigger.

Now, in terms of the macro level days, years between interruptions, this is one where, where we're far more philosophically relaxed.

It may well be that there is, there is no kind of profound sense in which these things are the same entity.

It's maybe a ship of theseus type, type concept.

And it's all linked together by some fairly weak long-term memory.

And are we really the same person as we were a week ago, a year ago?

We're far less philosophically concerned about that than we are by the micro level and the meso level.

But either way, the solutions are the classic personal identity solutions, which have been gone over plenty of times.

If, if you'll give us another five minutes before questions, we'd love to share some more kind of speculative thoughts with you.

And that's yeah, getting the thumbs up.

Wonderful. In which case, over to Andreas for some, for some fun thought-provoking stuff.

Well, it's just awesome. Awesome. Yeah, I mean, it's just kind of like this, yeah, deeper question and don't wait for me if my internet, yeah, runs out for a second.

But yeah, I mean, essentially, how does this influence, for example, yeah, beliefs of personal identity?

Next slide. I think once you start seeing the universe as a field and with maybe like fleeting topological pockets on it that actually very significantly informs these classic questions in philosophy, like who are you and do you continue to exist from one moment to the other?

And indeed, the other sheep of the Seas, like, is there any kind of like metaphysical, enduring ego, so to speak?

And I actually think that each of these views of personal identity could actually be reinterpreted in terms of claims about the field and field topology.

So something like closed individualism, this is the belief that you start existing when you're born, stop existing when you die, or, you know, maybe you go to heaven.

But the point is that you're kind of a separate self that endures over time that is different than other selves.

So here, we would want to find some kind of enduring sheath or enduring nested architecture that even though maybe each moment of awareness might be its own topological pocket, maybe there's some kind of higher order topological unity that does carry over a person's lifetime.

You know, that would be a possible way of ray finding in a in a actual, you know, significant physical way, the sense of identity over time.

Of course, completely speculative, it would would be a matter of actually finding it in the field.

But that's how you could, I think, still man this notion.

Next slide.

Then probably the most compatible and easy to justify view here would be empty individualism that every moment of experience is ontologically different.

And again, here that would be possible because we would have these fleeting hard boundaries in the space time continuum.

And as a consequence, you can look at the block universe and say, OK, there's a 40 pocket there, there's a 40 pocket over there.

And wherever you see a 40 pocket that is going to be a subject of experience.

And, you know, any actual continuity between moments of experience are either just causal or kind of illusory constructed with the memory module, as Chris was suggesting.

And then next slide.

And then finally, open individualism.

There's this view that we are all one consciousness.

We're all Krishna at the bottom bottom layer, so to speak.

Well, the way we could justify this within the topological account is that, well, actually, at the deepest topological level, we're all the field, which just happened to be on a given point in time,

segmentation of that field that prevents information propagation with other regions of the field.

So in this case, kind of the topological segmentation would actually be more of an epistemological boundary than it would actually be an ontological boundary.

It's like the idea that like deep down, actually, we are the field.

It's just that how the field knows itself is constrained by its topology, in which case, yeah, it's another way of still meaning we're all one consciousness in a very physically grounded way.

And yeah, I think I think that's it.

Unless I put anything else in the next slide.

Oh, yeah, yeah.

Just the last kind of like speculation here is, well, OK, if you are this topological pocket in the field, electromagnetic field, how can you, in a sense, take a look at that pocket and then recognize that?

And then reconstruct what the experience feels like.

And here, perhaps like a fun thought experiment that you could you could make is thinking about what it would be like to be an a photon trapped kind of within a hole of mirrors.

There's this article I wrote, yeah, 2018 about how to replicate a hyperbolic space using just double mirrors and gradient index optics.

And essentially within that structure that has those properties, a photon would have no way of telling the fact that it is not in hyperbolic space.

In other words, the index of refraction of a container would present itself as the actual geometry of the space from within.

So in a brain, presumably like the actual the actual way in which the local field potentials are stitched together and the relative index of refraction could potentially give rise to really arbitrary possible internal geometries where there's the illusion that you're in a space, whether it's in hyperbolic space, in spherical space, some irregular geometry.

All of those could be emulated with these with these power time and that may go a little bit a long way in actually explaining some of the most exotic experiences that people can have, which usually would challenge kind of theories of consciousness here.

Consciousness here, here there's a way of us teaching them together to explain them.

And I think, yeah, the last slide, I believe.

Yeah, so this is just illustrating that point.

I think I think this is it. Yeah.

Yeah, and we can leave this up on the screen.

This is just where we would go next with it.

But this seems like a good point to stop and hear your thoughts.

Awesome. Great presentations.

Well, Bernie, would you like to give a first reflection and I'll write down some questions and look at the live chat?

Yeah, sure.

I've been thinking a lot about similar stuff like this recently.

And when it's I the paper presents how the functional theories have the problem of arbitrariness.

So for example, if Markov blanket that can mark the blankets that could be the case when there's Markov blankets on the logical observers.

And I was thinking that maybe even if fine this with the Markov blanket paradigm is that the Markov black and that on the physical level is implemented as as using this topological pocket is the conscious Markov black.

So it's how you were referring the Markov black and solves the content problem, while the topological boundary solves all the other problems like object.

So it's not the object.

So it's not the object.

So it might be.

Maybe.

Yeah, no, that's something very quick to mention.

I mean, obviously we're in the active inference Institute podcast.

Well, absolutely one thing to mention is that our perspective is not in itself necessary or is not necessarily sufficient actually to satisfy the, you know, that we have specified.

However, we actually think it's very, very likely that topological pockets in the electromagnetic field do behave as Markov blankets.

And so it is in which you can actually interface these different paradigms where we would be saying that at the implementation level, you know, can then realize the minimization of free energy at the algorithmic level.

However, we would be saying, and maybe this would be a little bit of a twist or a non-trivial implication that if you were to simulate a, you know, hierarchically structured Markov blanket system in a computer, it could very well be from a certain abstraction be minimizing free energy, but it wouldn't have this.

It wouldn't be able to solve the boundary problem.

In other words, it is sort of like there is a significant overlap, but there are some slight places of divergence where you might say, Hey, a market blanket is actually not enough because it doesn't give you the implementation level explanation for how you get boundaries.

Chris, if you want to add anything.

Chris, if you want to add anything.

Yeah.

Yeah.

I mean, just to, just to agree with that.

I think this is the, the, if you go back to Mars three levels around kind of what substrate, uh, and we haven't discussed anything here about the algorithms that are actually taking place on or kind of within the brain that would ultimately get incorporated into the EM field.

Clearly you need that stuff as well.

None of this goes, makes any contribution to solving these more content problems of consciousness or how

we make inferences about the world, how we respond to those.

What it does say is that for, if there are algorithms that solve those problems and there are some good candidates out there, including yours, they need to be implemented or they need to reflect back to a field for it to become part of someone's unified.

Macro phenomenological experience.

If you did a, some sort of feed forward in silico version of this, or you did it, you implemented the, the algorithm using rocks, just lots and lots of rocks laid out forever.

Like that wonderful XKCD.

Um, example that, that is not going to create a bounds first person perspective anywhere.

Hmm.

Well, a lot to say about that.

Certainly Markov blankets.

And just to define it on a Bayesian graph.

So the map, not the territory, the Markov blanket is the set of nodes upon which two other sets of nodes are conditionally independent.

And that's a purely topological statement.

It doesn't matter how far, if there even is a spatial association or temporal association, it doesn't matter the distance, which is to say the geometry, but rather the blanket is a purely topological statement.

And that's what, among other things we've been exploring with Chris Fields in the physics as information processing.

And this question about the topology of communication for bounded observers.

And as we kind of alluded to with that content problem, the topology of communication is to use a geometric metaphor.

It's orthogonal to the content.

You could have two people on the phone sharing different things.

You could have a video file with different contents.

And so there, there's really a lot of layers that come into play.

And so I think from an instrumentalist view, I read your paper as a call for the introduction of topology into what is often only modeled as geometric.

A call for the integration of electromagnetism rather than a reliance on one or the other.

Of course, as introduced from the outset, a pluralism.

These are not exhaustive nor exclusive accounts.

And we don't know what the scope of the possible intelligences and awarenesses are.

So from a research perspective, if not beyond, why would we want to limit when we're not only interested in understanding the properties of any given system as we see it today?

Any thoughts or I can read a question in the chat?

Oh, that sounds good.

All right.

Turnkey Tyranny writes, how does topology show up phenomenologically?

Yeah, this is an excellent, excellent question.

I'm happy to take it.

Andres, do you want to go back to your notes?

You know, you've got that page of notes in there.

Sure, sure.

Let's do that.

Because this, this is because I've asked a very similar question of Andres.

And these sort of notes is what resulted in his attempts to explain it to me.

So let me take it.

Well, yes, very, this is very speculative.

And but but one wonderful resource, I think, I think, hopefully, I think it's wonderful, very worth trying out is we made a series of guided meditations about phenomenology.

It's not really kind of like for the sake of spirituality, it's more for the sake of exploring phenomenology with meditation.

The last one is called the phenomenology of ontology.

And that one helps you in a sense, get a sense of how beliefs about the reality.

Act, you read reality, of course.

Different to believe that you're in an electromagnetic field or you're in an electromagnetic pocket field.

You know, specific version of materialism or something like that.

So very different ontologies can give rise to very different feelings.

And so there's definitely the worry that like, well, if you think a lot about the electromagnetic field, aren't you maybe just interpreting your sensations within that paradigm?

But like, it's kind of like you're overfeeding on top of it.

But, you know, in a sense, I would argue that actually it's the other way around with something as generative and structured as electromagnetic field.

You can think in principle, explain the phenomenology of any ontology.

Like, why does it feel like, for example, that you can kind of like merge with the cosmos or something like that?

Why does it feel like you can become one with everything in meditation, let's say in various meditation states?

Well, with something like the topological explanation for the boundary problem, the way we might interpret that is as the vanishing of internal boundaries.

And so there is a significant difference between a global boundary, which is what we focused on, the overall kind of the macroscopic division that says whether you're inside or outside your experience.

But something that the electromagnetic field also affords and gives us as kind of part of its reach structure is the possibility of internal boundaries as well, which would be not boundaries that

that in, that wrap around a 4D object, but rather lower dimensional boundaries like meeting points or convergence points of the field lines within the pocket.

And so in this account, you know, whenever you have like the sensation of solidity, the feeling of, well, I am here, you're there, or there's a boundary between us.

That boundary would essentially be an internal boundary would be a region of the field where the field lines converge or the align.

But it's only along a certain cross section, meaning that, you know, actually your experience is still fully unified.

You can still go from one point to the other across a perceived barrier because they're still part of the whole experience.

They're still connected. In other words, with this kind of data structure, we can arrive at kind of like a lot of phenomenological details and trying to explain them from that perspective.

And I think it's going to be an enormous program in the long term, you know, involving physicists and mathematicians and exotic states of consciousness to say like, okay, that particular feature of experience that happens on such and such substance in such

meditation state, how does that map into the electromagnetic field?

And I think we will very likely find a lot of non-trivial connections with that sort of paradigm.

Chris, any thoughts on that?

Just to make the link to this, to some of the diagrams on the page, which is if you've got these non-4D closed boundaries, but other types of boundaries, the other types of shapes that Andrews was talking about, the drawings on the page are just examples of some of those.

But, and this is, as Andrew said, it's not really part of the solution, but it's part of imaginatively exploring where else it might go.

So if you've got feature binding, which is, you know, I'm seeing a cup and the cup is both blue and cup shaped.

That's feature binding. And then the phenomenological binding is my whole experience being unified.

You can imagine feature binding, putting the blueness and the cup is together.

That might be something that happens along one of these types of internal boundaries.

It might not, it might be that that can be done purely through informational associations where we're kind of agnostic on that.

But what's interesting about a 4D topology is within it, you can have lots of other complicated topologies provided they are not closed along the fourth dimension.

Cause then you've got, you sort of created another little mini bits of a little bit of awareness.

Oh, Bernie, go ahead. Yeah. I'll jump back to the other page.

Yeah. I wanted to say something else, not related to this exact question. So if you have ended it.

Maybe I'll, well, in decades, actually I'll, I'll mention one last thing about the, how electromagnetic fields may be related to phenomenology.

Uh, a very specific thing, uh, that happens on psychedelics and especially something like DMT is that you may end up having, uh, different regions of your experience have the exact same texture.

And have the feeling of synchronization between them.

And when that happens, they sort of become the same object, just like spatially distributed.

Um, the way I interpret this is whenever the exact same wave equation applies to two regions of your experience.

And waves can propagate from one region to the other that actually feels like they're the same object.

Yeah. One interesting thing you reflected in your literature review is when people think about quantum and cognitive systems, the ORC OR and other frameworks often are considering quantum as being electronic scale.

And therefore, a quantum theory of cognition or consciousness would entail some kind of relationship between the way that electrons and photons behave and then some difference that makes a difference at the biological level.

And in a way, what we've been exploring with Chris Fields and others is like a bypass where it's like, no, the quantum formalism, quantum topology, quantum information, simply and plainly describes cognitive processes where like order matters.

Now, there may or may not be effects associated with differences that make a difference at the electronic scale, but we don't need that in order to use like superposition and all of these different effects that empirically are the case in cognitive systems.

We're not waiting for some new mechanisms.

So, let's take it on.

So, one kind of like very awesome research lead, and then I'll explain what a research paradigm will look like. But essentially, first of all, like a very powerful lead is meditative cessations.

So, one of the most insane chapters in Mastering the Core Teachings of the Buddha by Daniel Ingram is the one called The Three Doors, where he describes the phenomenology of experiencing a peak meditation state.

That's called a cessation, where you blink out of reality for a second or well, for a sub-second moment, but you just blink out fully and then you get reconstituted.

And when you get reconstituted, it tends to be a very blissful, very rejuvenating type experience.

But the thing he highlights in that article is what it feels like to be really close to a cessation.

And it actually depends on what type of cessation it is.

There's roughly like three ways in which it can happen.

But the mind-blowing thing about that essay, that chapter, is he essentially describes different topologies of attention and awareness that may arise.

Where in these moments close to a cessation, he says, well, in one of those, you become kind of a torus and the torus flips the inside and outside and it twists around and then it cancels out and compresses and then you vanish.

And then there's like another variety where you become a cone and then all of your attention gets concentrated in the point of the cone and then that twists around and then vanishes.

And so there's like this reliable, you know, very, very exotic kind of like topological transformations that can happen to your attention and awareness.

And our guess would be that, well, a cessation, you know, an actual kind of full vanishing might be related to the topological pocket that your brain is generating, essentially opening up so that in some maybe non-trivial way,

the actual boundary between inside and outside gets dissolved for a fraction of a second, then it gets reconstituted.

But something really special happens in that moment that allows for deep insight or something along the lines of relieving a lot of internal stress, maybe like a coronal mass ejection.

There might be something going on at the topological level.

So now in terms of, yeah, kind of like further, you know, more sophisticated research paradigms, the thing that we're looking into, there's a lot more to be unpacked and developed here, but it's what kind of electromagnetic resonance modes that can supervene on particular graphs?

You know, whether we're talking about the connectome or whether we're talking about, you know, the geometric eigenmodes, more recent work where you model the entire cortex as a geometric surface.

And then you apply neural field theory to that.

Well, so like what kind, like all of that works so far, essentially what it focuses on is oscillations in the electric field without modeling the magnetic field.

So I'm actually, you know, I would suspect that many of the connectome harmonics or eigenmode, geometric eigenmodes may be physically unrealizable simply because the magnetic field lines may collide with each other.

So like some of those may not be possible.

But one of the crazy fascinating, unusual predictions that may come out of our whole paradigm is that once we have a really good mapping of what resonant modes arise in the brain on different states of consciousness, we would predict that A, when you're fully awake, normally conscious, there will be a very central big topological loop inside you.

So that whenever you're, let's say, that whenever you're, let's say, in deep sleep, you might see a kind of mind dust, you might see lots of tiny little pockets.

And maybe when you're in an ecstatic, you know, experience like deep meditation, it may do something else.

Like the particular resonant modes there may create open field lines that don't form a loop.

So that is like the sort of crazy, very novel type of prediction or at least research paradigm that this enables.

That's, that's really cool.

Chris, anything out of that?

Yeah, just to draw the link between the fourth bullet here and what Andres is setting out.

I don't think any of this is, there are some toy bits of modeling you could do that could increase confidence one way or another.

But in terms of an actual experimental prediction, you would need to combine this with some theory of where, which bits of the brain are doing this.

So if Ward and Guevara are right, that it's the thalamus that's doing this stuff, we would expect to see around the thalamus a closed pocket.

And it should be possible in principle, like this is probably billions of pounds worth of effort, but to model the thalamus and the surrounding bits enough.

And you can find out if there would actually be a closed pocket.

If there would be one in theory, you can see roughly where it would be.

Obviously it moves around a bit, but it's normally around here.

You then should be able to, I don't know if ethical consent is even possible for something like this, but disrupt that with a very, very targeted externally M source.

Think, you know, transcranial stimulation, but just, you know, laser precise.

If we've now broken the bit around the thalamus, but left everything else more or less untouched, we predict if

Ward and Guevara plus us is correct.

We predict that your first personal awareness would blink out like deep sleep, like anesthesia, like a meditative cessation.

If we're wrong or the combination of us is wrong, then you'll still be conscious, but you might just experience something really bloody weird.

But you're still experience, your contents have changed, not the, not the actual container of consciousness hasn't blinked in and out.

And then you could apply that to other bits of the brain, wherever you think there's, I mean, our other, other prediction is that it must be enclosing the immediate memory model.

But we're unsure exactly where that is.

That is also an outstanding research question.

So this is, I mean, it feels like doable in our lifetimes, I think, but none of this is, you know, give us 10 grand and we'll do it this summer.

None of it is that easy.

Hmm.

Yeah.

Very interesting.

And neuroimaging already utilizes electromagnetic measurements.

And there are things that currently are described within a biomolecular paradigm, like, for example, changes in effective connectivity between different regions of the brain.

Well, there's already a tract, let's just say, between certain brain regions.

And so maybe if there's, if the action potentials traveling down that 100 lane highway are aligned, then it's going to be like this rippling EM continuity.

Whereas if they were totally disarrayed along that tract, then you wouldn't have those two regions contained.

But it'd be like kind of like two regions, like blowing smoke rings at each other if they had enough synchrony to push through the trouble and the friction such that there was a pocket that was able to, you know, you squirt hot water into a cold pool.

There's ways to angle it and do it in all of the combinations of viscosity that it just diffuses instantly.

There's also ways where there would be a high temperature toroidal form that could pass through a colder media because it maintains a topological boundary.

A lot of these things are, again, described anatomically or molecularly or electronically.

But the magnetic, the duality between electronics and magnetism is not contentious.

So there may be a lot of phenomena that could be revisited.

And I don't know whether that would provide indirect evidence with what we already know or suggest new plausible experiments.

But it's an approach.

Yeah, welcome.

And thanks, Bernie, as well, for the pointers towards Chris Fields and Michael Levin.

I may come back to you for the QFT on memory formation in case I've spotted some stuff on that, but I'm not sure I've spotted the one it sounded like you were referring to.

So, yeah, it's an exciting time for this sort of research.

Yeah, one last connection there with active inference is you have the electronic and the magnetic fields influencing each other.

That's the basis of what philosophers might consider in terms of this epiphenomenalism or not question.

And if we had an active inference situation of two songbirds, the active states of one become the sensory of the other.

So if they were just passing messages every single time step to each other, then they would both be engaged in their own blanketed perception on the inbound and action selection on the outbound.

And so they could basically be in this continual oscillatory relationship with causal efficacy both ways.

Again, or you could design a situation where the causal efficacy only flowed one way.

But the point is, this is a topological approach that lets us explore that in a way that isn't only appealing to molecular or material phenomena.

And then leaving the whole question, which is the whole question of the informational and the closure and the binding,

leaving all those to be resolved by slicing and dicing with a scalpel.

Which is not even a viable option as far as we already understand.

Bernie, and then the authors, what are your closing thoughts?

Or how would you suggest that people get in the game?

Or what should they do with their own topologies?

Maybe I could note that Andres could expand on if he wants to know synodity of violence and neuro any link.

Well, yeah, I mean, I guess the he or I, I mean, STB introduced by Mike in 2015 on Principia Qualia, but the correspondence between symmetry of an hour.

Yeah.

You know, it's been like,

up to TRI for many years now, you know, you can see a 2020 presentation about it.

And there's many more content about it.

But the main idea, the way it would actually, you know, work out here is the regions of the topologically closed field that behave like a smooth field.

They potentially correspond to a very blissful experience.

So, you know, I'm very, very optimistic of the potential of, okay, like, something like a peak blissful experience, peak 5MEO DMT experience or something like that.

But it would probably, I suspect, have like a very internally simplified topology where, you know, the geometry is as smooth as possible.

So that's kind of like an interesting, yeah, prediction or angle on all of these.

I think, yeah, maybe to wrap up, I would say that we, you know, would enormously welcome anybody who is technically savvy for like doing interesting simulations of the electromagnetic field.

I mean, we are, we're doing ourselves some like really interesting experiments and in combination with classical neural networks.

But anybody's kind of like, you know, the intersection between machine learning, electromagnetic fields and

simulations and, you know, statistics, machine learning.

Essentially, that kind of intersection is something that could really potentially accelerate this research.

You know, if you had the technical competency to, yeah, sort of like work on the analysis I was describing, finding the electromagnetic resonant modes of a graph and then analyzing its topology, you know, we would love to collaborate.

That's like the sort of content we, we dream of producing and yeah, in the next year or two.

So thank you.

And thank you for having us.

It's been fantastic.

Really, really appreciate it.

I appreciate that.

Chris?

Nothing substantial to add.

Just, just, just my thanks as well.

And, you know, I've, I've scribbled down some thoughts about, about the Markov blankets and about some of those different research areas as well.

So, you know, we're getting stuff from this as well.

So yeah, and hopefully this helps reach a few new people who might come up with either a brilliant idea why something we've missed and this, this just can't work for some, we want to hear that as well.

Um, but even more constructively ways that this could be tested or built out more gradually modeled.

Um, yeah.

Thank you for having us.

Cool.

Cool.

New ways of thinking within even one topological frame enables new action selection in the area and the era of video chatting.

We've made new topological relationships, new connectivities that then disperse yet again.

And so it continues.

Thank you all though.

Thank you so much.

Till next time.

Very lovely.

Thank you.

Amazing stuff.

See ya.

Bye.

Bye.

Bye.

Thank you.